

Trial Transcript Day Two: Temperature Excursion and Causation

October 7, 2025 — proceedings resumed

THE COURT: Good morning. Members of the jury, continue to keep an open mind. Yesterday you heard contract evidence. Today the parties will address the physical chronology and the asserted relationship between the environmental event and Blue Ember's disposition.

MS. HOLT: Blue Ember briefly recalls Adrian Locke to complete the foundation for Granite Heron's investigation.

MR. LOCKE: I remain under oath.

Q. Did Granite Heron open an internal deviation after learning the curve?

A. Yes. Operations, quality, maintenance, and dispatch records were collected under Tess Holloway's coordination.

Q. What two principal system failures were identified?

A. Standby power was not energized and verified, and the relevant telemetry exception did not reach the active response queue.

Q. Did the investigation accuse Blue Ember of causing either failure?

A. No.

Q. Did it decide product potency?

A. No. It documented transport performance and left scientific disposition to Blue Ember.

Q. Did the report conceal the missed active queue?

A. No. It identified routing as a contributing failure and preserved the configuration evidence available for review.

Q. Was the source logger export retained?

A. Yes. A readable graph was later created from it, and the relationship is documented.

Q. Did Granite Heron smooth away the maximum?

A. No. The report states the threshold crossings and minus 42.8-degree maximum.

Q. Were later interviews distinguished from event-time entries?

A. Yes. The investigation labels its sources and approximations.

Q. What corrective measures followed?

A. Two-person power confirmation, live alert-profile testing, revised yard checks, and focused retraining.

Q. Did those measures mean the company conceded Blue Ember's dollar demand?

A. No. We corrected the system while reserving questions about cause and loss.

Q. Do you dispute that a procedure failure occurred within Granite Heron's control?

A. No.

MS. HOLT: We offer the carrier investigation subject to the limiting instruction that factual source entries and later analysis remain distinguishable.

MR. MERCER: No objection under that limitation.

THE COURT: Admitted. The jury may consider how and when each portion was created.

Q. Was any factual correction made without retaining the original entry?

A. No. Later explanation remained separately dated and attributable.

Examination of Tess Holloway by Blue Ember

MS. HOLT: Blue Ember calls Tess Holloway adversely.

CLERK: Ms. Holloway, do you swear that your testimony today will be truthful?

MS. HOLLOWAY: I do.

Q. What position did you hold on September 12, 2024?

A. Transport Operations Manager for Granite Heron's Fletcher controlled-temperature facility.

Q. Were you responsible for implementing the BE-214 operating procedure?

A. Yes, including training, staging controls, escalation, and investigation. I did not personally insert the cable.

Q. What did the procedure require before tractor shutdown?

A. Personnel had to connect a compatible source, energize it, and observe that the refrigeration controller remained powered and cooling.

Q. Was inserting a plug sufficient?

A. No. Positive verification was a separate critical step.

Q. Did the temperature requirement continue in the yard?

A. Yes. Staging did not suspend custody or the specified band.

Q. Could a closed trailer be assumed cold without checking?

A. No. The controller and monitoring systems existed because visual inspection was inadequate.

Q. Did your written role include authority to suspend the temperature band?

A. No. I could manage operations and emergencies, but I could not amend Blue Ember's release limit.

Equipment readiness

Q. Identify the vehicle combination.

A. Tractor GH-4821 and trailer TR-7714.

Q. Was the trailer generally capable of frozen service?

A. Yes. Pretrip and maintenance records did not show a critical defect before the route.

Q. Did the later investigation find fuel depletion?

A. No.

Q. Was the standby connector broken?

A. Post-event testing did not establish a hardware break. The source was not energized, and the state was not verified.

Q. Did the facility have qualified backup frozen space?

A. Yes, subject to a controlled transfer. The load was not moved there during the event.

Q. What target appeared on the route sheet?

A. Minus seventy degrees Celsius, with the allowed range from minus eighty through minus sixty.

Q. Did you see an approved grace period?

A. No.

Q. Was CTX9-44721 identified in the file?

A. Yes, as the CryoTrace payload monitor.

Q. Did the pretrip check alone guarantee later continuity?

A. No. It established initial readiness. Personnel still had to operate the equipment correctly at every power transition.

Q. Was standby use an unusual emergency step?

A. No. It was a planned part of the Fletcher staging process and therefore should have been performed under the checklist.

Fletcher arrival and power change

Q. When did the vehicle enter the Fletcher facility?

A. At 12:46 Eastern Daylight Time.

Q. When did it reach bay F-4?

A. 12:49.

Q. Describe the 12:52 entry.

A. It records that the standby cable was seated.

Q. What did the investigation find about the breaker?

A. It was not moved into the energized position.

Q. When was the tractor turned off?

A. At approximately 12:53.

Q. Is there a contemporaneous entry confirming active refrigeration after that?

A. No. The required positive check is absent.

Q. Did Blue Ember direct the stop?

A. No. Granite Heron planned and controlled it.

Q. Was the sealed payload transferred to a different custodian?

A. No. It remained inside our trailer.

Q. What happened when tractor power ended without standby?

A. The refrigeration unit stopped operating, and the controlled space began to warm.

Q. Was anyone watching a successful post-shutdown controller display?

A. The investigation found no such observation. If it had occurred, the absence of power should have been apparent.

Q. Did the cable entry record electrical current?

A. No. It recorded a physical step and did not prove that the source was live.

Threshold and telemetry

Q. When did the payload record exceed minus sixty?

A. At 13:08.

Q. What does the 13:10 system entry show?

A. A telemetry exception was directed to an inactive archive queue instead of the current response queue.

Q. Did an active dispatcher acknowledge it then?

A. No.

Q. Was the source information erased?

A. No. The device retained the readings. The routing failure delayed human response.

Q. Could a local check independently reveal the condition?

A. Yes. An unpowered controller display would show that refrigeration was not operating.

Q. Was a local check made at the threshold crossing?

A. Not at 13:08.

Q. Would the expected procedure have produced a Blue Ember call then?

A. Personnel should have investigated immediately and begun the escalation path.

Q. Did passive storage in an archive satisfy that purpose?

A. No. The protocol required action, not merely later retrievability.

Q. Was the archived exception intentionally ignored?

A. There is no evidence of intent. The profile was configured incorrectly, and the system did not place the message before active personnel.

Q. Would a live test have revealed the problem?

A. Yes, which is why testing became part of corrective action.

Discovery and restart

Q. How was the stopped unit found?

A. A yard check at 13:36 observed the dark controller.

Q. What did personnel do?

A. They restored tractor power and restarted refrigeration at 13:38, then kept the load closed while cooling resumed.

Q. What maximum did the payload logger record?

A. Minus 42.8 degrees at 13:37.

Q. Were you already at the bay then?

A. No. I was notified at 13:41 and arrived around 13:45.

Q. What did you direct?

A. Confirmation of active cooling, preservation of records, attempts to reach Blue Ember, and construction of an event chronology.

Q. Why was the shipper not opened?

A. Opening could add heat and break controls. With refrigeration restored, keeping the sealed trailer closed was judged protective.

Q. Did that response make the previous condition compliant?

A. No. It limited further exposure.

Q. Did you alter the maximum because cooling had begun?

A. No. The full curve remained in the file.

Q. Was the 13:37 maximum inferred from an interview?

A. No. It came from the retained CTX-9 time series.

Recovery, notice, and delivery

Q. When did the CTX-9 return to the authorized band?

A. At 14:02.

Q. How long was it above the upper boundary?

A. Fifty-four minutes.

Q. What contact efforts appear?

A. Calls to Dr. Solis's route at 13:49 and 13:58, followed by contact with Blue Ember receiving at 14:07.

Q. Was that immediate notice from 13:08?

A. No. It occurred after discovery and restart.

Q. When did the truck leave Fletcher?

A. At 14:18.

Q. Why was delivery continued?

A. Blue Ember had frozen receiving capacity and disposition authority. After recovery, direct delivery was considered safer than an unnecessary transfer.

Q. When did the destination sign physical receipt?

A. 15:26.

Q. Did that signature approve the excursion?

A. No. Blue Ember marked the lot for quarantine and review.

Q. Did your team tell receiving that the material was unquestionably usable?

A. No. We reported an exception and restored cooling; disposition remained open.

Q. Was delivery delayed so Granite Heron could test the lot itself?

A. No. We did not possess the scientific authority or laboratory function for that decision.

Q. Did the later drive add another warm interval?

A. The designated record stayed within the band after recovery through delivery.

Records and root-cause analysis

Q. Which records supported your chronology?

A. The route entries, custody sheet, CTX-9 export, controller information, contact log, interviews, and maintenance context.

Q. Were all times reconstructed from memory?

A. No. System and event-time records supplied most of them. Interviews filled context and were labeled as later sources.

Q. What root cause did the company adopt?

A. An unverified, unenergized power transition combined with ineffective alert routing and insufficient local detection.

Q. Was “operator error” the entire analysis?

A. No. We addressed procedure design, supervisory checks, profile control, and training as system contributors.

Q. Did you identify an origin packaging defect?

A. No evidence tied the temperature curve to an origin defect.

Q. Did receiving cause the warm interval?

A. No. Recovery occurred before the trailer left our facility.

Q. Can you determine whether the lot remained valid scientifically?

A. No. That is beyond my transport role.

Q. Did the investigation distinguish immediate cause from broader contributors?

A. Yes. Loss of power was immediate, while checklist design, verification, queue configuration, and delayed detection contributed to duration.

Q. Was the chronology adopted before your deposition?

A. Yes. It appears in the ordinary event investigation.

Corrective actions

Q. What changed in the power procedure?

A. Two people must now confirm the energized source and active controller for this service class before tractor shutdown.

Q. What changed in telemetry?

A. Specialized profiles receive a live test, and selection of an inactive queue triggers a warning and supervisor review.

Q. What changed in the yard?

A. Frozen-trailer observation intervals were shortened and documented. A dark or abnormal controller requires immediate escalation.

Q. Were assigned personnel retrained?

A. Yes, with practical demonstration and retained completion evidence.

Q. Did the company inspect trailer TR-7714?

A. Yes. It did not find a condition requiring permanent retirement.

Q. Why take corrective action if you dispute damages?

A. Operational improvement should occur regardless of a later legal disagreement over product effect or money.

Q. Does the action file have owners and dates?

A. Yes. It tracks completion and effectiveness rather than relying on informal assurance.

MS. HOLT: No further questions at this stage.

THE COURT: The jury may consider corrective measures only for the limited purpose allowed by the earlier ruling, including explanation of system operation where relevant, and not as automatic proof of legal liability.

Examination of Holloway by Granite Heron

MS. WEBB: Ms. Holloway, was the trailer incapable of deep-frozen service?

MS. HOLLOWAY: No. The event came from the transition and response failures we discussed.

Q. Was the payload compartment opened during warming?

A. No authorized opening appears.

Q. Did the shipper break or spill?

A. Not according to the custody record.

Q. Once the controller was discovered, did personnel act?

A. Yes. Restart occurred at 13:38, and cooling continued without opening the doors.

Q. Was the value back within specification before the route resumed?

A. Yes, at 14:02, with departure at 14:18.

Q. Did anyone intend to harm Blue Ember's material?

A. No.

Q. Did your investigation calculate 4.2 million dollars?

A. No. It was an operations investigation.

Q. Did it conclude that every vial lost activity?

A. No. We made no such scientific finding.

Q. Did the operations report state that recovery was irrelevant?

A. No. It documented recovery as part of exposure duration while leaving significance to qualified product witnesses.

Q. Was any exterior package defect observed when the doors later opened?

A. No such defect appears in the delivery record.

Q. Does the controller measure the same location as the payload logger?

A. Not necessarily. The records can corroborate mechanism and timing, while the designated CTX-9 represents the payload measurement used here.

Q. Can you say the product core itself reached minus 42.8 degrees?

A. I can state what the designated logger reported at its qualified location. Product-core interpretation requires different expertise.

Q. Did you decide that Blue Ember had to destroy the lot?

A. No.

Q. Did Granite Heron prevent testing?

A. No. The lot was delivered for Blue Ember's quarantine and assessment.

Q. Could prompt notice have changed the readings already recorded?

A. It could not reverse time, but it could have provided earlier consultation and protective choices.

Q. Did recovery matter at all?

A. It mattered for limiting exposure and scientific assessment. It did not mean the excursion never happened.

Q. Have you told the jury what damages to award?

A. No. My testimony is confined to transport facts and company response.

MS. WEBB: Thank you.

Dr. Solis recalled — receiving and quarantine

MS. HOLT: Dr. Solis, when did Blue Ember receiving learn of a confirmed exception?

DR. SOLIS: The contact log shows receiving reached at 14:07. Full device review occurred around delivery.

Q. What happened at 15:26?

A. Staff accepted physical custody into frozen quarantine, checked seals, preserved the shipper, and began the data download.

Q. Was the lot placed into ordinary use?

A. No. It remained segregated.

Q. What internal identifier was opened?

A. Deviation BE-QA-DEV-2024-0912.

Q. What did the first review establish?

A. The monitor had been above the upper limit for fifty-four minutes and reached a temperature far warmer than the accepted band.

Q. Did receiving claim visible spoilage?

A. No. The exterior was intact. Temperature and intended-use questions are not resolved by appearance.

Q. Did Blue Ember preserve material for testing?

A. Yes. Quarantine allowed controlled sampling and investigation.

Q. Was frozen storage maintained during that review?

A. Yes. We prevented an additional condition from confusing the original event and kept custody traceable for each sample.

Q. Did receiving personnel decide the legal claim amount?

A. No. They protected the lot and created quality records.

Device and custody evidence

Q. How did Blue Ember confirm the logger identity?

A. The serial on the custody documents matched CTX9-44721 in the export and receiving record.

Q. Was its clock basis known?

A. Yes. The shipment records used Eastern Daylight Time.

Q. Did Blue Ember move timestamps to fit the route?

A. No. The source sequence was preserved, and the transport entries were compared without rewriting it.

Q. What did the curve look like?

A. A progressive warming followed by recovery after refrigeration restarted, not a single isolated point.

Q. Was the raw export itself shown to the jury as a legal conclusion?

A. No. The readable exhibit identifies data and annotations. Witnesses explain what each element represents.

Q. Did the shipment remain with Granite Heron throughout the warm interval?

A. Yes, according to the stipulated custody record.

Q. Had Blue Ember touched the load at 13:37?

A. No. It was in Fletcher.

MS. HOLT: We offer the custody-and-logger exhibit.

THE COURT: Admitted under the agreed foundation and limiting instruction.

Confirmatory assay

Q. Why did Blue Ember commission confirmatory testing?

A. To gather product-impact information rather than assume the outcome from temperature alone.

Q. What did tested samples show?

A. They retained measurable biological activity under the fictional method.

Q. Did Blue Ember report total inactivity?

A. No.

Q. What did the test not answer?

A. It did not examine every vial, prove uniformity, reconstruct continuous custody, or establish that a lot exposed beyond the validation limit could support the intended process claim.

Q. Was activity irrelevant?

A. It was relevant and considered. It was not the only release attribute.

Q. Did the method establish a safe commercial dose?

A. No. There was no commercial or patient use here.

Q. Could additional testing eliminate every uncertainty?

A. Not without consuming material and still leaving the historical custody deviation. The team weighed benefit, time, and what validation required.

Q. Was the assay record created before litigation?

A. Yes, during the September quality investigation.

Q. Did counsel choose which tested values went into the original report?

A. No. The laboratory documented its results through the ordinary controlled process.

September 16 disposition

Q. What conclusion did the quality unit reach?

A. Lot BE214-VL-2409A would not be released for its intended validation use.

Q. On what date?

A. September 16, 2024.

Q. State the rationale accurately.

A. The magnitude and duration exceeded the no-tolerance criterion, and limited activity testing could not demonstrate the controlled, uniform history required for this validation lot.

Q. Did the disposition call Granite Heron fraudulent?

A. No. It addressed the material, not intent.

Q. Were alternatives considered?

A. Additional testing, limited research use, and replacement were discussed. None restored the lot's validation function.

Q. Was destruction immediate?

A. No. The lot stayed segregated while evidence and claim needs were preserved, then was destroyed under controlled procedure later.

Q. Did any research use create a material financial credit?

A. No quantifiable benefit offsetting the claimed embodied cost was identified.

Q. Was the decision documented with signatures?

A. Yes, in the deviation and lot-disposition file.

Q. Did that file acknowledge the assay's retained activity?

A. Yes. The stated basis was not total inactivity; it was the validation significance of the documented exposure and remaining uncertainty.

Causal sequence

Q. Would this lot have been rejected for the same custody reason without the recorded excursion?

A. No. The no-tolerance criterion would have been satisfied if the shipment stayed within range.

Q. Was another preexisting lot defect identified?

A. No defect requiring the same rejection had been identified before transport.

Q. Did Blue Ember manufacture the power-transition failure?

A. No.

Q. Was replacing a validation lot a foreseeable response to a failed validation condition?

A. Yes. The documents disclosed the use and required strict records for that reason.

Q. Does that answer the dollar amount?

A. No. Costs and schedule effects require separate financial proof.

Q. Could later site selection affect the delay component?

A. Yes. That is the mitigation dispute. It does not change why the original lot entered replacement status.

Q. Are original loss and later mitigation the same question?

A. They occur in sequence and can be evaluated separately.

Q. Did Blue Ember treat the assay as though it showed nothing?

A. No. The written rationale states what it did and did not establish.

Cross-examination on disposition

MR. MERCER: Blue Ember wrote the specification, correct?

DR. SOLIS: It owned the product criterion, and Granite Heron accepted it.

Q. Could Blue Ember have written an allowed excursion?

A. It could have if scientifically justified before the shipment. Revision 3 did not include one.

Q. Was the decision reviewed by an outside regulator?

A. No. This was internal validation governance.

Q. Did you ask Cobalt Finch or Aurora Quay whether nonvalidation research use had value?

A. The team considered internal and program uses. Those facilities were evaluated for replacement manufacture, not purchase of exposed material.

Q. Did the assay produce a numerical failure?

A. The activity result itself was not a categorical failure. The custody attribute was.

Q. Are those two attributes given equal economic weight automatically?

A. No. Intended use determines materiality.

Q. Could a jury disagree with your business judgment?

A. It can assess whether the consequence was reasonable and caused under the Court's instructions.

Q. Are you asking jurors to accept a conclusion because of your title alone?

A. No. They should examine the contemporaneous rationale and underlying evidence.

Q. Blue Ember controlled when to stop testing?

A. Yes, through a documented quality process.

Q. More tests might have produced more information?

A. Possibly, but not a compliant historical record. Testing consumes time and material and must answer a useful question.

Q. You chose complete replacement?

A. For validation, yes.

Q. Was the lot used to make any commercial sale?

A. No.

Q. Then the 1.35 million dollars is a cost figure, not a sales price?

A. Correct.

Q. And the 2 million delay category arises from later economic expectations?

A. It arises from a bounded schedule effect. Brooks will explain it.

Q. Could choosing a March campaign enlarge that period?

A. The chosen replacement timing affects it, which is why the October alternative is before the jury.

Q. Did Granite Heron choose March 3?

A. No. Blue Ember made that decision after evaluating options.

Q. So at least part of causation depends on post-event reasonableness?

A. For avoidable delay, yes; not for the temperature event or original disposition itself.

Redirect on quality causation

MS. HOLT: Did Blue Ember create Revision 3 after seeing the excursion?

DR. SOLIS: No. It was effective August 1.

Q. Did Granite Heron accept it before pickup?

A. Yes.

Q. Would retained activity authorize rewriting a validation condition after the fact?

A. No. It informed assessment but did not erase the agreed criterion.

Q. Did Blue Ember claim a retail value for the lot?

A. No. The original-lot amount is traced cost.

Q. Is a challenge to later delay the same as proof that the initial lot remained usable?

A. No. Granite Heron may contest both, but they require different evidence.

Q. Did you approach the disposition as a damages advocate?

A. No. The record was created in September to decide what to do with material before a complaint was filed.

Q. Why preserve that timing?

A. It shows the decision was tied to quality information then available, not invented after counsel developed a case.

Q. Was the product already the subject of a filed lawsuit on September 16?

A. No. The civil complaint came later.

Exhibits, limiting instruction, and adjournment

THE COURT: The assay summary and lot-disposition exhibit are admitted with a limiting instruction. They show the tests performed, recorded result, and Blue Ember's documented decision. They do not bind the jury on legal causation or damages.

MR. MERCER: Granite Heron renews its objection to any suggestion that quality authority makes the economic consequence conclusive.

THE COURT: The objection is preserved and the limiting instruction addresses it. The jury will decide weight.

MS. HOLT: Blue Ember asks that the Court also remind the jury that retained activity does not make the temperature criterion disappear.

THE COURT: Agreed. Both points will be included in the final charge.

Members of the jury, tomorrow you will hear evidence about an offered replacement slot and the parties' mitigation positions. Do not use that later evidence to prejudge the transport chronology, and do not assume that an admitted operational failure determines every loss. Continue to avoid discussion and outside research.

CLERK: All rise. Court is recessed until October 8 at 9:00 a.m.